
Dylan Ballard

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SUMMARY

Full-stack software engineer specializing in React 19, TypeScript, Python, and cloud infrastructure. Engineered LLM document extraction pipelines on AWS SageMaker achieving 99% accuracy across 10K+ pages/day. Builds production systems end-to-end — from PostgreSQL schema design and FastAPI backends to component-driven React UIs — with a focus on performance, scalability, and developer experience. Currently pursuing M.S. Computer Science at George Mason University.

EXPERIENCE

Eigennet

2024 – Present

Software Engineer

Reston, VA

- Engineered LLM document extraction pipeline on AWS SageMaker achieving **99% accuracy**, processing **10K+ pages/day** and reducing manual review time by **85%**
- Automated Python SFTP delivery pipeline — metadata packaging, zip creation, encrypted transfer — delivering **500+ classified documents monthly** with **99.7% success rate**, eliminating **20+ hours/week** of manual processing
- Implemented EC2/S3 batch workflows with API-driven transmission tracking, reducing delivery errors by **90%** across production pipelines
- Designed Django validation system detecting schema drift before delivery, preventing downstream data failures across partner systems

Freddie Mac

2023

Software Engineering Intern

McLean, VA

- Designed Java-based data validation tool detecting anomalies and schema drift across large financial CSV datasets, reducing manual reconciliation effort by **60%**
- Implemented automated version-mismatch alert system, catching **3x more** data integrity issues before production deployment
- Accelerated quarterly reconciliation cycle from **2 weeks to 3 days** through automated CSV comparison workflows

PROJECTS

Embark

embark-theta.vercel.app

React, TypeScript, FastAPI, Python, Supabase, Vercel

- Developed full-stack productivity platform with RPG mechanics serving **50+ active users** — XP progression, quest engine, rarity-tiered item collection, and real-time leaderboards
- Architected event-driven backend using FastAPI and Supabase real-time subscriptions with WebSocket endpoints for live quest progress updates

Cinetik

cinetik.app

React, TypeScript, FastAPI, Python, PostgreSQL, TanStack Query, Zustand, Tailwind CSS

- Developed full-stack media tracking platform integrating the TMDb API with TanStack Query and Zustand state management across **500K+ catalog records**
- Optimized PostgreSQL query patterns and indexing strategy achieving sub-100ms response times with a responsive Tailwind CSS interface

React 19, TypeScript, Electron, Python, FastAPI, Supabase, Vite

- Developed cross-platform notes and planning application for desktop (Electron) and web using a shared React 19 codebase in an npm workspaces monorepo
- Implemented domain-first layered architecture with shared TypeScript packages, CodeMirror markdown editing, and FastAPI/Supabase backend

Shipkitty

github.com/dballard10

React, TypeScript, Hono, Bun, Drizzle ORM, PostgreSQL (Neon), shadcn/ui

- Architected AI agent observability dashboard for monitoring and replaying autonomous agent sessions with real-time action tracking and undo support
- Implemented Hono API server on Bun runtime with Drizzle ORM, PostgreSQL (Neon), and GitHub OAuth authentication

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL, Java, HTML, CSS

Frontend: React 19, Next.js 16, React Native, Vite, Tailwind CSS v4, Zustand, TanStack Query, shadcn/ui, Radix UI, Framer Motion, Three.js

Backend: FastAPI, Hono, Django, Node.js, Bun, Pydantic, REST APIs, GraphQL, WebSockets, SSE Streaming, Auth (JWT, OAuth), Celery, asyncio

Databases & ORM: PostgreSQL, MySQL, SQLite, Supabase, Redis, Drizzle ORM, Prisma

AI & Automation: LLM Integrations (OpenAI, Anthropic, Bedrock), Multi-Agent Architectures, Prompt Engineering, Playwright, SageMaker, ETL Pipelines, Pandas, NumPy

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, SageMaker), Docker, Terraform, Vercel, Fly.io, Railway, GitHub Actions, CI/CD, Linux, Shell Scripting

Tools & Testing: Git, Electron, Vitest, Jest, pytest, @testing-library/react, Zod, npm Workspaces, Postman, Figma, Jira, Agile/Scrum

EDUCATION

George Mason University

2024 – Present

Master of Science, Computer Science

James Madison University

2020 – 2024

Bachelor of Science, Computer Science